

# AHMED MOHAMED KOTB

## Backend Developer

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### PROFESSIONAL SUMMARY

Backend developer focused on building typed, well-structured REST APIs with NestJS and TypeScript over PostgreSQL and TypeORM, with additional experience in Node.js / Express, MongoDB, and ASP.NET Core with Entity Framework Core. Practised in relational data modeling, JWT authentication and role-based authorization with Passport, DTO-level request validation, and layered design using Clean Architecture, CQRS, and Mediator. Also build real-time services over WebSocket / Socket.IO and consume my own APIs from React, so contracts are designed with the client in mind. ICPC regional-qualification competitor with strong data structures and algorithms fundamentals.

### PROJECTS

#### Team Task Management Board — REST API

NestJS · PostgreSQL · 2026

*NestJS · TypeScript · PostgreSQL · TypeORM · Passport JWT · class-validator · class-transformer · React (client)*

- Designed and built a NestJS + PostgreSQL REST API for team task management, organized into feature modules with services, controllers, and providers.
- Modeled a normalized relational schema in TypeORM (users, teams, boards, tasks) with explicit relations and cascade behavior.
- Implemented JWT authentication with Passport strategies and guard-based role authorization separating owner, manager, and member permissions.
- Enforced request validation and response shaping end to end with class-validator and class-transformer DTOs, and consumed the API from a React + TanStack Query client.

#### E-commerce Clothing Platform — Web API

ASP.NET Core · 2024

*ASP.NET Core Web API · Entity Framework Core · SQL Server · FluentValidation · React (client)*

- Built the ASP.NET Core Web API powering an e-commerce platform, structured with Clean Architecture across domain, application, and infrastructure layers.
- Implemented ASP.NET Identity, JWT access tokens with refresh-token rotation, and request validation with FluentValidation.
- Applied CQRS with the Mediator pattern to separate read and write paths, and exposed server-side product filtering, sorting, and pagination.

#### Real-time Chat Service

Node.js · 2023

*Node.js · Express · MongoDB · Socket.IO*

- Built an Express + MongoDB backend with a Socket.IO layer handling bidirectional messaging, room membership, and message persistence.

#### Education Platform — API & Deployment

Node.js · 2022

*Node.js · Express · MongoDB · JWT · AWS Lightsail*

- Developed the Node.js / Express API and MongoDB data layer for an online education platform, secured with JWT authentication and deployed to AWS Lightsail.

#### Intelligent CCTV Security System

Graduation Project · 2024

*Python · Deep Learning · WebSocket · Multithreading*

- Built the real-time processing backend for an AI security system detecting violence, weapons, intrusion, and faces, using multithreaded frame handling and WebSocket-based alerting.

### TECHNICAL SKILLS

**Backend:** NestJS, Node.js, Express, ASP.NET Core Web API, REST API design, WebSocket & Socket.IO

**Databases & ORMs:** PostgreSQL, SQL Server, MongoDB, TypeORM, Entity Framework Core, relational data modeling, SQL

**Auth & Validation:** Passport JWT, role-based authorization, guards, ASP.NET Identity, refresh tokens, class-validator, class-transformer, FluentValidation

**Architecture:** Clean Architecture, CQRS, Mediator, layered and modular design, OOP, Clean Code, Data Structures & Algorithms

**Programming Languages:** TypeScript, JavaScript, C#, Python, SQL

**Frontend (working knowledge):** React, TypeScript, TanStack Query, React Hook Form, Zod, Tailwind CSS, Axios

**AI-Assisted Development:** Claude, Claude Code, GitHub Copilot, ChatGPT — prompt engineering, AI-assisted code review, debugging, test generation

**Tools & Practices:** Git, GitHub, AWS Lightsail, Agile methodology, teamwork

### EDUCATION

**B.Sc. in Computer Science — Faculty of Computers and Information, Suez University**

2020 – 2024

Overall Grade: Very Good | Cumulative GPA: 3.25 / 4.00

## COURSES & CERTIFICATIONS

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- Nest.js Developer Lab 2026: Build a Robust API with Authentication, Articles, and User Profiles — LinkedIn Learning, 2026
- The Complete Node.js Course: RESTful Web Services with Node.js, Express & MongoDB — MaharaTech / ITI MOOC, 2022
- TypeScript Fundamentals · Introduction to MongoDB — MaharaTech / ITI MOOC, 2022
- Advanced Full-Stack Web Development Nanodegree — Udacity, 2022 · C#, .NET Core API, Entity Framework & SQL Server

## ACTIVITIES & ACHIEVEMENTS

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- **ICPC ECPC Qualifications, 2023** — represented Suez University in a team-based collegiate programming contest, solving complex algorithmic problems under time constraints.

## LANGUAGES

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**Arabic:** Native

**English:** Professional working proficiency (speaking, listening, reading)